

35.6 The Tragedy of the Commons

We have argued above that if property rights are well defined, there will be no problem with production externalities. But if property rights are not well defined, the outcome of the economic interactions will undoubtedly involve inefficiencies.

In this section we will examine a particularly well-known inefficiency called “the tragedy of the commons.”⁵ We will pose this problem in the original context of a common grazing land, although there are many other possible illustrations.

Consider an agricultural village in which the villagers graze their cows on a common field. We want to compare two allocation mechanisms: the first is the private ownership solution where someone owns the field and decides how many cows should graze there; the second is the solution where the field is owned in common by the villagers and access to it is free and unrestricted.

Suppose that it costs a dollars to buy a cow. How much milk the cow produces will depend on how many other cows are grazed on the common land. We'll let $f(c)$ be the value of the milk produced if there are c cows grazed on the common. Thus the value of the milk per cow is just the average product, $f(c)/c$.

⁴ Anna Oberthur, “Almond Growers Face Need for Bees,” Associated Press, February 29, 2004.

⁵ See G. Hardin, “The Tragedy of the Commons,” *Science*, 1968, 1243–47.

How many cows would be grazed on the common if we wanted to maximize the total wealth of the village? In order to maximize the total amount of wealth, we set up the following problem:

$$\max_c f(c) - ac.$$

It should be clear by now that the maximal production will occur when the marginal product of a cow equals its cost, a :

$$MP(c^*) = a.$$

If the marginal product of a cow were greater than a , it would pay to put another cow on the commons; and if it were less than a , it would pay to take one off.

If the common grazing ground were owned by someone who could restrict access to it, this is indeed the solution that would result. For in this case, the owner of the grazing grounds would purchase just the right amount of cows to maximize his profits.

Now what would happen if the individual villagers decided whether or not to use the common field? Each villager has a choice of grazing a cow or not grazing one, and it will be profitable to graze a cow as long as the output generated by the cow is greater than the cost of a cow. Suppose that there are c cows currently being grazed, so that the current output per cow is $f(c)/c$. When a villager contemplates adding a cow, the total output will be $f(c+1)$, and the total number of cows will be $c+1$. Thus the revenue that the cow generates for the villager will be $f(c+1)/(c+1)$. He must compare this revenue to the cost of the cow, a . If $f(c+1)/(c+1) > a$, it is profitable to add the cow since the value of the output exceeds the cost. Hence the villagers will choose to graze cows until the average product of a cow is driven to a . It follows that the total number of cows grazed will be $\hat{c}$, where

$$\frac{f(\hat{c})}{\hat{c}} = a.$$

Another way to derive this result is to appeal to free entry. If it is profitable to graze a cow on the common field, villagers will purchase cows. They will stop adding cows to the common only when the profits have been driven to zero, that is, when

$$f(\hat{c}) - a\hat{c} = 0,$$

which is just a rearrangement of the condition in the last paragraph.

When an individual decides whether or not to purchase a cow, he looks at the extra value he will get $f(c)/c$ and compares this to the cost of the cow, a . This is fine for him, but what has been left out of this calculation is the fact that his extra cow will reduce the output of milk from all the *other*

cows. Since he is ignoring this **social cost** of his purchase, too many cows will be grazed on the common ground. (We assume that each individual has a number of cows that is negligible relative to the total number grazed on the common.)

This argument is illustrated in Figure 35.4. Here we have depicted a falling average product curve, since it is reasonable to suppose that the output per cow declines as more and more cows are grazed on the common land.

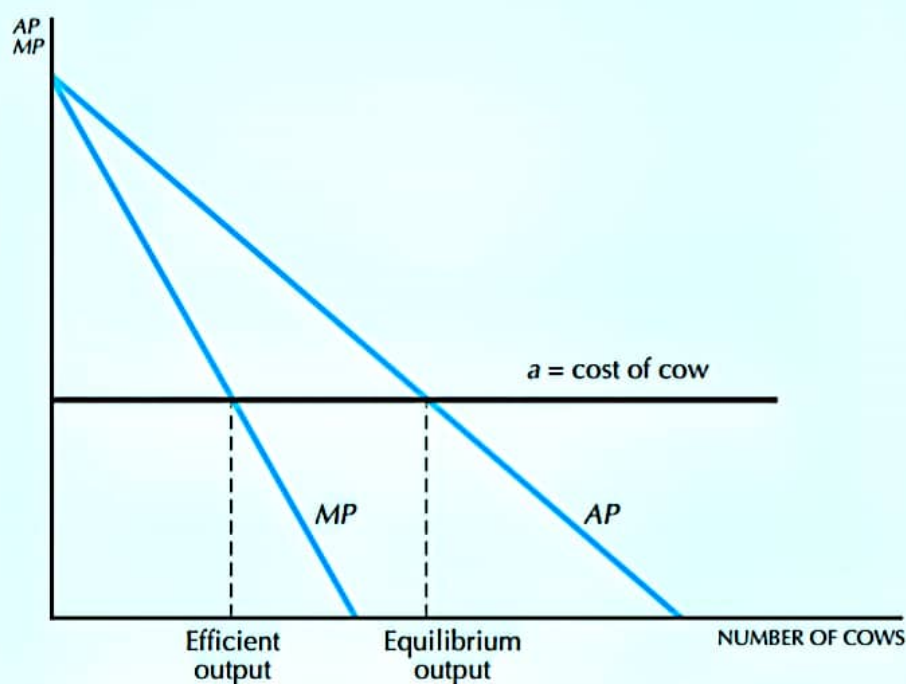


Figure 35.4

The tragedy of the commons. If the grazing area is privately owned, the number of cows will be chosen so that the marginal product of a cow equals its cost. But if grazing area is common property, cows will be grazed until the profits are driven to zero; thus the area will be overgrazed.

Since the average product is falling, it must be that the marginal product curve always lies below the average product curve. Thus the number of cows where the marginal product equals a must be less than where the average product equals a . The field will be overgrazed in the absence of a mechanism to restrict use.

Private property provides such a mechanism. Indeed, we have seen that if everything that people care about is owned by someone who can control its use and, in particular, can exclude others from overusing it, then there are by definition no externalities. The market solution leads to a Pareto

efficient outcome. Inefficiencies can only result from situations where there is no way to exclude others from using something, a topic that we will investigate in the next chapter.

Of course, private property is not the only social institution that can encourage efficient use of resources. For example, rules could be formulated about how many cows can be grazed on the village common. If there is a legal system to enforce those rules, this may be a cost-effective solution to providing an efficient use of the common resource. However, in situations where the law is ambiguous or nonexistent, the tragedy of the commons can easily arise. Overfishing in international waters and the extermination of several species of animals due to overhunting are sobering examples of this phenomenon.